

Question 1

A permutation of a set is an arrangement of its members into a sequence. If the set is $\{1, 2, 3\}$, then the possible permutations are: $(1, 2, 3)$, $(1, 3, 2)$, $(2, 1, 3)$, $(2, 3, 1)$, $(3, 1, 2)$, and $(3, 2, 1)$. Hence the number of permutations of this set is 6.

a) Find a recurrence relation for the number of permutations of a set with n elements. Let $P(n)$ be the number of permutations of a set with n elements.

- $P(0) = 1$, since there is still one way to arrange an empty set.
- $P(1) = 1$
- $P(2) = 2 \times 1 = 2$
- $P(3) = 3 \times 2 \times 1 = 6$
- $P(4) = 4 \times 6 \times 2 \times 1 = 24$

This suggests a recurrence relation of the form:

$$\begin{cases} P(n) = n \times P(n-1) & \text{if } n > 0 \\ P(0) = 1 & \text{if } n = 0 \end{cases}$$

b) Use this recurrence relation to find the number of permutations of a set with n elements using iteration. $P(n) =$
 $n \times P(n-1)$

$$= n \times (n-1) \times P(n-2)$$

$$= n \times (n-1) \times (n-2) \times P(n-3)$$

$$= \dots$$

$$= n \times (n-1) \times (n-2) \times \dots \times 2 \times 1 \times P(0)$$

Substitute the base case into the expanded relation:

$$P(n) = n \times (n-1) \times (n-2) \times \dots \times 2 \times 1 \times 1$$

$$= n! \implies \text{A set with } n \text{ elements has } n! \text{ permutations.}$$

Question 2

Find a recurrence relation to compute the number of bit strings. Write down a procedure that takes n as an input parameter and returns the number of bit strings. You can provide the pseudocode or implement it in your favorite programming language.

```
fn count_bit_strings(n: u32) -> u32 {  
  if n == 0 {  
    // Base case: There is 1 bit string of length 0, which is an empty string  
    return 1;  
  } else {  
    return 2 * count_bit_strings(n - 1);  
  }  
}
```

Question 3

How can you modify the procedure so that instead of returning the number of bit strings, it returns a list of all bit strings? Please provide the procedure/program.

```
fn gen_bit_strings(n: u32, current: String, result: &mut Vec<String>) {  
    if n == 0 {  
        result.push(current);  
    } else {  
        gen_bit_strings(n - 1, current.clone() + "0", result);  
        gen_bit_strings(n - 1, current.clone() + "1", result);  
    }  
}
```

Question 4

How can you verify given a bit string whether it contains two consecutive zeros? Please provide a procedure/program.

```
fn has_consecutive_zeros(s: &str) -> bool {  
    if s.len() < 2 {  
        return false;  
    }  
    let bytes = s.as_bytes();  
    if bytes[0] == b'0' && bytes[1] == b'0' {  
        return true;  
    }  
    has_consecutive_zeros(&s[1..])  
}
```

Question 5

a) Find a recurrence relation for the number of bit strings of length n that contain a pair of

consecutive 0s. Let b_n be the number of bit strings of length n that contain two consecutive 0s.

- If a bit string of length n ends in 00, the first $n - 2$ bits can be any valid bit string, which gives 2^{n-2} possibilities.
- If a bit string of length n ends in 01, the first $n - 3$ bits can be any valid bit string, which gives b_{n-1} possibilities.
- If a bit string of length n ends in 10, the first $n - 4$ bits can be any valid bit string, which gives b_{n-2} possibilities.

Therefore, the recurrence relation is:

$$b_n = 2^{n-2} + b_{n-1} + b_{n-2}$$

b) What are the initial conditions? $b_0 = 0$, since an empty bit string cannot contain two consecutive 0s.

$b_1 = 0$, since a bit string of length 1 cannot contain two consecutive 0s.

$b_2 = 1$, since there is only one bit string of length 2 that contains two consecutive 0s, which is 00.

c) How many bit strings of length seven contain two consecutive 0s? Using the recurrence relation:

$$b_7 = 2^{7-2} + b_6 + b_5$$

$$b_7 = 2^5 + b_6 + b_5$$

$$b_7 = 32 + b_6 + b_5$$

$$b_7 = 32 + (2^{6-2} + b_5 + b_4) + b_5$$

$$b_7 = 32 + 16 + b_5 + b_5 + b_4$$

$$b_7 = 48 + 2b_5 + b_4$$

$$b_7 = 48 + 2(2^{5-2} + b_4 + b_3) + b_4$$

$$b_7 = 48 + 8 + b_4 + b_4 + b_3$$

$$b_7 = 56 + 2b_4 + b_3$$

$$b_7 = 56 + 2(2^{4-2} + b_3 + b_2) + b_3$$

$$b_7 = 56 + 8 + b_3 + b_3 + b_2$$

$$b_7 = 64 + 2b_3 + b_2$$

$$b_7 = 64 + 2(2^{3-2} + b_2 + b_1) + b_2$$

$$b_7 = 64 + 4 + b_2 + b_2 + b_1$$

$$b_7 = 68 + 2b_2 + b_1$$

$$b_7 = 68 + 2(2^{2-2} + b_1 + b_0) + b_1$$

$$b_7 = 68 + 2 + b_1 + b_1 + b_0$$

$$b_7 = 70 + 2b_1 + b_0$$

$$b_7 = 70 + 2(1) + 0$$

$$b_7 = 72$$

Question 6

a) Find a recurrence relation for the number of bit strings of length n that contain three consecutive 0s. Let c_n be the number of bit strings of length n that contain three consecutive 0s.

- If a bit string of length n ends in 000, the first $n - 3$ bits can be any valid bit string, which gives 2^{n-3} possibilities.
- If a bit string of length n ends in 001, the first $n - 4$ bits can be any valid bit string, which gives c_{n-1} possibilities.
- If a bit string of length n ends in 010, the first $n - 5$ bits can be any valid bit string, which gives c_{n-2} possibilities.
- If a bit string of length n ends in 100, the first $n - 6$ bits can be any valid bit string, which gives c_{n-3} possibilities.

Therefore, the recurrence relation is:

$$c_n = 2^{n-3} + c_{n-1} + c_{n-2} + c_{n-3}$$

b) What are the initial conditions? $c_0 = 0$, since an empty bit string cannot contain three consecutive 0s.

$c_1 = 0$, since a bit string of length 1 cannot contain three consecutive 0s.

$c_2 = 0$, since a bit string of length 2 cannot contain three consecutive 0s.

$c_3 = 1$, since there is only one bit string of length 3 that contains three consecutive 0s, which is 000.

c) How many bit strings of length seven contain three consecutive 0s? Using the recurrence relation:

$$c_7 = 2^{7-3} + c_6 + c_5 + c_4$$

$$c_7 = 2^4 + c_6 + c_5 + c_4$$

$$c_7 = 16 + c_6 + c_5 + c_4$$

$$c_7 = 16 + (2^{6-3} + c_5 + c_4 + c_3) + c_5 + c_4$$

$$c_7 = 16 + 8 + c_5 + c_4 + c_5 + c_4 + c_3$$

$$c_7 = 24 + 2c_5 + 2c_4 + c_3$$

$$c_7 = 24 + 2(2^{5-3} + c_4 + c_3 + c_2) + 2c_4 + c_3$$

$$c_7 = 24 + 4 + c_4 + c_3 + c_4 + c_3 + c_2$$

$$c_7 = 28 + 2c_4 + 2c_3 + c_2$$

$$c_7 = 28 + 2(2^{4-3} + c_3 + c_2 + c_1) + 2c_3 + c_2$$

$$c_7 = 28 + 4 + c_3 + c_2 + c_3 + c_2 + c_1$$

$$c_7 = 32 + 2c_3 + 2c_2 + c_1$$

$$c_7 = 32 + 2(2^{3-3} + c_2 + c_1 + c_0) + 2c_2 + c_1$$

$$c_7 = 32 + 2 + c_2 + c_1 + c_2 + c_1 + c_0$$

$$c_7 = 34 + 2c_2 + c_1$$

$$c_7 = 34 + 2(2^{2-3} + c_1 + c_0) + c_1$$

$$c_7 = 34 + 2 + c_1 + c_1 + c_0$$

$$c_7 = 36 + 2c_1 + c_0$$

$$c_7 = 36 + 2(1) + 0$$

$$c_7 = 38$$

The number of bit strings of length seven that contain three consecutive 0s is 38.

Question 7

Solve $a_n = 5a_{n-1} - 6a_{n-2} - 2$ for $n \geq 2$, $a_0 = 1$, $a_1 = 0$.

Given the recurrence relation $a_n = 5a_{n-1} - 6a_{n-2}$, we can solve it using the characteristic equation method.

The characteristic equation is $r^2 = 5r - 6$, which simplifies to $r^2 - 5r + 6 = 0$.

Factoring the quadratic equation, we get $(r - 2)(r - 3) = 0$.

Therefore, the roots are $r_1 = 2$ and $r_2 = 3$.

The general solution is $a_n = c_1(2^n) + c_2(3^n)$.

Using the initial conditions $a_0 = 1$ and $a_1 = 0$:

$$a_0 = c_1(2^0) + c_2(3^0) = c_1 + c_2 = 1$$

$$a_1 = c_1(2^1) + c_2(3^1) = 2c_1 + 3c_2 = 0$$

Solving the system of equations:

$$c_1 + c_2 = 1$$

$$2c_1 + 3c_2 = 0$$

$$c_1 = 3$$

$$c_2 = -2$$

Therefore, the solution to the recurrence relation is $a_n = 3(2^n) - 2(3^n)$.

Question 8

Solve $a_n = 4a_{n-2} - 2$ for $n \geq 2$, $a_0 = 0$, $a_1 = 4$.

Given the recurrence relation $a_n = 4a_{n-2}$, we can solve it using the characteristic equation method.

The characteristic equation is $r^2 = 4$, which simplifies to $r^2 - 4 = 0$.

Factoring the quadratic equation, we get $(r - 2)(r + 2) = 0$.

Therefore, the roots are $r_1 = 2$ and $r_2 = -2$.

The general solution is $a_n = c_1(2^n) + c_2(-2^n)$.

Using the initial conditions $a_0 = 0$ and $a_1 = 4$:

$$a_0 = c_1(2^0) + c_2(-2^0) = c_1 - c_2 = 0$$

$$a_1 = c_1(2^1) + c_2(-2^1) = 2c_1 - 2c_2 = 4$$

Solving the system of equations:

$$c_1 - c_2 = 0$$

$$2c_1 - 2c_2 = 4$$

$$c_1 = 2$$

$$c_2 = 2$$

Therefore, the solution to the recurrence relation is $a_n = 2(2^n) + 2(-2^n)$, which simplifies to $a_n = 2^{n+1}$.

Question 9

Solve $a_n = \frac{a_{n-2}}{4}$ for $n \geq 2$, $a_0 = 1$, $a_1 = 0$.

Given the recurrence relation $a_n = \frac{a_{n-2}}{4}$, we can solve it using the characteristic equation method.

The characteristic equation is $r^2 = \frac{1}{4}$, which simplifies to $r^2 - \frac{1}{4} = 0$.

Factoring the quadratic equation, we get $(r - \frac{1}{2})(r + \frac{1}{2}) = 0$.

Therefore, the roots are $r_1 = \frac{1}{2}$ and $r_2 = -\frac{1}{2}$.

The general solution is $a_n = c_1(\frac{1}{2})^n + c_2(-\frac{1}{2})^n$.

Using the initial conditions $a_0 = 1$ and $a_1 = 0$:

$$a_0 = c_1(\frac{1}{2})^0 + c_2(-\frac{1}{2})^0 = c_1 + c_2 = 1$$

$$a_1 = c_1(\frac{1}{2})^1 + c_2(-\frac{1}{2})^1 = \frac{1}{2}c_1 - \frac{1}{2}c_2 = 0$$

Solving the system of equations:

$$c_1 + c_2 = 1$$

$$\frac{1}{2}c_1 - \frac{1}{2}c_2 = 0$$

$$c_1 = \frac{1}{2}$$

$$c_2 = \frac{1}{2}$$

Therefore, the solution to the recurrence relation is $a_n = \frac{1}{2}(\frac{1}{2})^n + \frac{1}{2}(-\frac{1}{2})^n$, which simplifies to $a_n = \frac{1}{2^{n+1}}$.

Question 10

Solve $a_n = a_{n-1} + 6a_{n-2}$ for $n \geq 2$, $a_0 = 3$, $a_1 = 6$

Given the recurrence relation $a_n = a_{n-1} + 6a_{n-2}$, we can solve it using the characteristic equation method.

The characteristic equation is $r^2 = r + 6$, which simplifies to $r^2 - r - 6 = 0$.

Factoring the quadratic equation, we get $(r - 3)(r + 2) = 0$.

Therefore, the roots are $r_1 = 3$ and $r_2 = -2$.

The general solution is $a_n = c_1(3^n) + c_2(-2^n)$.

Using the initial conditions $a_0 = 3$ and $a_1 = 6$:

$$a_0 = c_1(3^0) + c_2(-2^0) = c_1 - c_2 = 3$$

$$a_1 = c_1(3^1) + c_2(-2^1) = 3c_1 - 2c_2 = 6$$

Solving the system of equations:

$$c_1 - c_2 = 3$$

$$3c_1 - 2c_2 = 6$$

$$c_1 = 0$$

$$c_2 = -3$$

Question 11

Solve $a_n = 7a_{n-1} - 10a_{n-2}$ for $n \geq 2$, $a_0 = 2$, $a_1 = 1$.

Given the recurrence relation $a_n = 7a_{n-1} - 10a_{n-2}$, we can solve it using the characteristic equation method.

The characteristic equation is $r^2 = 7r - 10$, which simplifies to $r^2 - 7r + 10 = 0$.

Factoring the quadratic equation, we get $(r - 5)(r - 2) = 0$.

Therefore, the roots are $r_1 = 5$ and $r_2 = 2$.

The general solution is $a_n = c_1(5^n) + c_2(2^n)$.

Using the initial conditions $a_0 = 2$ and $a_1 = 1$:

$$a_0 = c_1(5^0) + c_2(2^0) = c_1 + c_2 = 2$$

$$a_1 = c_1(5^1) + c_2(2^1) = 5c_1 + 2c_2 = 1$$

Solving the system of equations:

$$c_1 + c_2 = 2$$

$$5c_1 + 2c_2 = 1$$

$$c_1 = -1$$

$$c_2 = 3$$

Therefore, the solution to the recurrence relation is $a_n = -1(5^n) + 3(2^n)$, which simplifies to $a_n = 3(2^n) - 5^n$.

Question 12

For integer multiplication, we solved the n -bit problem by using three solutions of $n/2$ -bit problems and additional c_n operations. However, the analysis was very complicated. If we use a simpler analysis and solve the n -bit problem by using four solutions of $n/2$ -bit problems and additional $c_1 n$ operations, then what will be the recurrence? What will be the solution to that recurrence achieved from the simpler analysis?

If we solve an n -bit integer multiplication problem using four subproblems of size $\frac{n}{2}$ and additional $c_1 n$ operations, the recurrence relation becomes:

$$T(n) = 4T(n/2) + c_1 n$$

This is a general form of the Master Theorem, where $a = 4$, $b = 2$, and $f(n) = c_1 n$, where a represents the number of subproblems, b represents the factor by which the problem size is reduced, and $f(n)$ represents the cost of dividing the problem and combining the solutions.

The solution to this recurrence relation is:

$$T(n) = \Theta(n^{\log_b a}) = \Theta(n^{\log_2 4}) = \Theta(n^2)$$

Therefore, the time complexity of this simpler analysis is $\Theta(n^2)$.

This simpler analysis indicates that using four subproblems instead of three still results in a time complexity of $\Theta(n^2)$ for integer multiplication.

The dominating term in the time complexity is the recursion depth.

Question 13

Suppose that $f(n) = f(\frac{n}{3}) + 1$ when n is a positive integer divisible by 3, and $f(1) = 1$. Find

a) $f(3)$. $f(3) = f(\frac{3}{3}) + 1 = (1) + 1 = 2$

b) $f(27)$. $f(27) = f(\frac{27}{3}) + 1 = (f(9) + 1) + 1 = ((f(3) + 1) + 1) + 1 = (((1) + 1) + 1) + 1 = 4$

c) $f(729)$.

- $3 = 3^1 \implies f(3^1) = 2$
- $9 = 3^2 \implies f(3^2) = 3$
- $27 = 3^3 \implies f(3^3) = 4$
- $81 = 3^4 \implies f(3^4) = 5$

Thus, for any $n = 3^k$, $f(n) = k + 1 \implies f(n) = \log_3 n + 1 \implies$
 $f(729) = \log_3 729 + 1 = 6$

Question 14

Find $f(n)$ when $n = 2^k$, where f satisfies the recurrence relation $f(n) = f(\frac{n}{2}) + 1$ with $f(1) = 1$.

- $f(1) = 1$
- $f(2) = f(\frac{2}{2}) + 1 = f(1) + 1 = 1 + 1 = 2$
- $f(4) = f(\frac{4}{2}) + 1 = f(2) + 1 = 2 + 1 = 3$
- $f(8) = f(\frac{8}{2}) + 1 = f(4) + 1 = 3 + 1 = 4$
- $f(16) = f(\frac{16}{2}) + 1 = f(8) + 1 = 4 + 1 = 5$

From the pattern, we can see that for any $n = 2^k$, $f(n) = k + 1 \implies f(n) = \log_2 n + 1$.

Question 15

Find $f(n)$ when $n = 3^k$, where f satisfies the recurrence relation $f(n) = 2f(\frac{n}{3}) + 4$ with $f(1) = 1$

- $f(1) = 1$
- $f(3) = 2f(\frac{3}{3}) + 4 = 2f(1) + 4 = 2 + 4 = 6$
- $f(9) = 2f(\frac{9}{3}) + 4 = 2f(3) + 4 = 2(6) + 4 = 16$
- $f(27) = 2f(\frac{27}{3}) + 4 = 2f(9) + 4 = 2(16) + 4 = 36$

From the pattern, we can see that $f(3^k) = 2^{k+1} + 2^{k+2} = 3 \times 2^{k+1} = 3 \times 2^{\log_3 n + 1} = 3n^{\log_3 2 + 1}$.